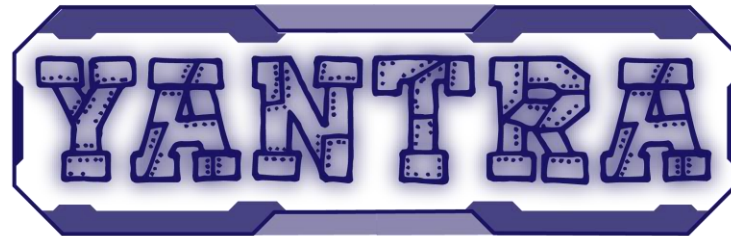


Medical and bio-additive manufacturing:



Scaffolds for Tissue Engineering

- Tissue engineering has been used to replace failing or malfunctioning organs such as skin, liver, pancreas, heart valve leaflet, ligaments, cartilage and bone. This has given rise to the interests in applying RP techniques to build scaffolds either to induce surrounding tissue and cell in-growth or serve as temporary scaffolds for transplanted cells to attach and grow onto. These scaffolds can be designed in three dimensions on CAD taking into consideration the porosity and good interconnectivity for tissue induction to occur. The function of cells, such as in bone and cartilage regeneration, is dependent on the three-dimensional spatial relationships. As such, the geometry of these hard tissues are critical to its function [38–41]. RP has been able to lend itself to producing complex geometry scaffolds

Inter-Vertebral Spacers

- Human spinal vertebrae can **disintegrate** due to conditions such as osteoporosis or **extreme forces acting** on the spine.
- In the management of such situations, a spacer is usually required as part of the spinal fixation process.
- **RP** has been investigated for the production of such spacers as it is an **ideal process** to fabricate 3D structures with good interconnecting pores for the promotion of tissue in-growth.
- Other considerations for producing such an implant are that the **material is biocompatible**, and that the **mechanical compressive strength** of the spacer is able to withstand spinal loads.

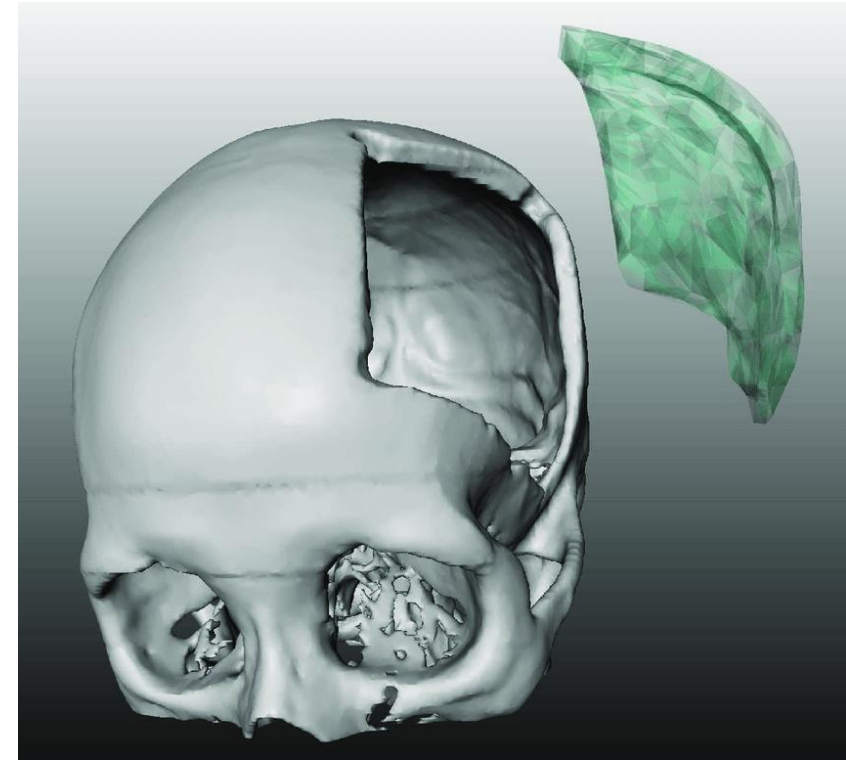


- A process, developed at the Nanyang Technological University for such a purpose, uses a solid **RP master pattern** of the spacer to **produce a soft mold**.
- Stainless steel bearings coated with a formulated binder system are then cast into the soft mold under vacuum.
- Upon curing, the part is ejected from the mold.
- The part then undergoes debinding and sintering processes to produce the final part.
- The primary advantage of this process is its ability to use a solid RP pattern to produce from this a porous structure with controllable pore sizes and mechanical strengths



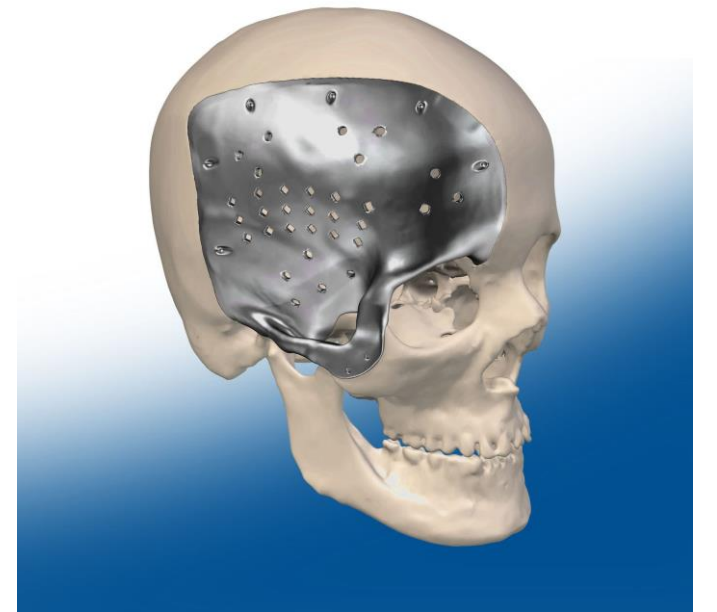
Cranium Implant

- A patient suffered from a large frontal cranium defect after complications from a previous meningioma tumor surgery.
- This left the patient with a missing cranial section, which caused the geometry of the head to look deformed.
- Conventionally, a titanium-mesh plate would be hand-formed during the operation by the surgeon. This often resulted in inaccuracies and time spent for trial and error.



computerized tomography (CT)

- Using RP, standard preparations of the patient were made and a **computed tomography (CT)** scan of the affected area and surrounding regions was taken during the pre-operation stage.
- The three-dimensional CT data file was transferred to a CAD system and the missing section of the cranium topography was generated.
- After some **software repair** and **cleaning up** were carried out on the newly generated section, an inverted mold was produced on CAD.



- This three-dimensional solid model of the mold was saved in *.STL format and transferred to the RP system, such as the SLS, for building the mold.
- The SLS mold was produced and used to mechanically press the titanium-mesh plate to the required three-dimensional profile of the missing cranium section.
- During the operation, the surgeon cleared the scalp tissue of the defect area and fixated the perfectly pre-profiled plate onto the cranium using self-tapping screws.
- The scalp tissue was then replaced and sutured. At post-operation recovery, results observed showed improved surgical results, reduced operation time and a reduce probability of complications.

Computer-aided tissue engineering (CATE) advances modeling, design and fabrication of tissue scaffolds

For example, CATE can apply biomimetic design approach to introduce multiple biological and biophysical requirements into the scaffold design

CATE can also integrate both biomimetic and nonbiomimetic features into the scaffold modeling database to form high fidelity and smart scaffolds.

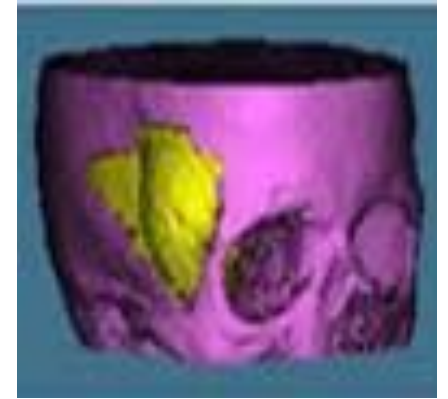
Biomimetic features can be based upon real anatomical data regenerated from CT/MRI images, or can be created purely within a CAD environment

- for example, channels and porous structures.

Non-biomimetic features do not imitate nature but can be designed as drug storage chambers, mechanical elements, and attachment interfaces for tubes, sensors, electronics, and other devices..

In the following hypothetical case, we present how a seeded implant would be designed and used for repairing a skull defect after a tumor had been removed.

To create the external structure of the implant we took the CT data, and used the 3D-reconstruction tools in MIMICS to isolate the tumor. The STL file for the skull and the STL file for the tumor were then exported from MIMICS and imported into Geomagic. Within Geomagic we used a mirroring operation to mirror the tumor from the right side of the skull to the left side, i.e. the healthy side. We then used the Boolean intersection between the tumor and skull to generate a bone implant structure.



- We then mirrored the implant back across the plane to complete the final external architecture of the implant Through this procedure, we essentially replaced the void created by the removal of the tumor,
- with a mirrored copy of the normal bone structure. Using this method, we were able to create a complex structure that matched the symmetry of the patient's skull.